

Tiancheng (Alex) Zhao

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EDUCATION

Hong Kong University of Science and Technology <i>PhD Student, Computer Science and Engineering</i>	Hong Kong SAR 2027/01 - 2029/12 (<i>Expected</i>)
Georgia Institute of Technology <i>M.S Computer Science (Specialization: Computer System & High Performance Computing)</i>	Atlanta, GA, USA 2024/08 - 2026/08
Shandong University <i>B.E Computer Science, GPA:91.96, Honored Bachelor Degree(Top 2%)</i>	Jinan, China 2019/09 - 2023/06
The University of Hong Kong <i>Exchange Student, B.E of Computer Science</i>	Hong Kong SAR 2022 Spring

EXPERIENCES

High Performance Computing Group, Georgia Institute of Technology <i>MS Student, Thesis Advisor: Assistant Prof. Helen Xu</i>	2024/08 – Present Atlanta, GA
• pdGRASS: Parallel Dynamic Graph Sparsification Engine. Built a multi-core optimized dynamic graph sparsification system with emphasis on throughput, cache locality, and parallel scheduling efficiency. Designed batch-based rebalancing and locality-aware partitioning to reduce synchronization overhead and improve scalability under heavy update workloads. Demonstrated strong scaling and low-latency updates on large-scale graph datasets.	
Tencent, WeChat HPC Group <i>Research & Development Intern, High Performance Computing</i>	2026/05 – 2026/08 Shenzhen, China
• Co-designed system-level optimizations for on-device small models, leveraging mobile GPU acceleration and efficient memory management for low-latency inference.	
A*Star Agency for Science, Technology and Research of Singapore <i>Student Researcher, Supervisor: Dr. Xin He</i>	2025/08 – 2026/01 Singapore
• Authored a comprehensive survey that systematizes long-term memory mechanisms for LLM-based agents from a unified representation–management perspective. Proposed a coherent taxonomy across token-level, intermediate latent, and parameter-level memory, and organized existing approaches via a shared construction–update–query abstraction to clarify system trade-offs and guide future LLM memory system design.	
Xpeng <i>Software Development Engineer Intern</i>	2025/06 – 2025/08 Santa Clara, CA
• Built and benchmarked TensorRT- and TorchScript-based inference pipelines for autonomous driving perception models, conducting systematic accuracy and latency comparisons across deployment configurations. • Profiled end-to-end inference workflows to identify computational hot spots and developed high-performance image processing kernels to reduce latency and improve throughput. • Analyzed and optimized system-level logging and monitoring pipelines to reduce runtime overhead and alleviate production system load during large-scale inference.	
Automated Reasoning & Transformation of Software Group, HKUST <i>Research Postgraduate Student, Supervisor: Assistant Prof. Jiasi Shen</i>	2023/04 – 2024/08 Hong Kong, SAR
• Built a runtime-driven code synthesis system that infers UI control flow and navigation from black-box Android apps and generates production-ready cross-platform code via automated IR construction and model-assisted inference. Enabled scalable full-app regeneration with high code fidelity and deployment readiness. • Contributed to OSVBench, a benchmark for evaluating LLMs on operating system specification generation and verification-oriented code synthesis. Assisted in large-scale evaluation and analysis of model performance on long-context, verification-grade tasks, identifying reliability and consistency gaps in real-world OS development workflows.	
High Performance Computing Lab, Shandong University <i>RA, Advisor: Prof. Weiguo Liu</i>	2021/09 – 2023/06 Jinan, China
• Participated in ASC22, PAC22 and IPCC22 parallel computing and supercomputer competitions as undergraduate team leader (got together 45K RMB grants). • Familiar with parallel computing optimization methods such as cache optimization, pipeline rescheduling, etc and classic parallel models like OpenMP, Pthreads and MPI. Experiences in Intel SYCL, Data Parallel C++, CUDA. • Final Year Project: Parallel Iterator for Unstructured Grids on Multi-core Platform.	
Intelligence in Healthcare Group, The Chinese University of Hong Kong <i>RA, Mentor: Assistant Prof. Yu Li</i>	2022/02 – 2022/07 Remote
• Participate in a Bioinformatics Program on enzyme encoding. Implement baseline experiments with multiple ML Models like SVM, LR, Adaboost, etc and accomplish training task.	
Institute of High Energy Physics, Chinese Academy of Sciences <i>Intern, Calculation Center, Supervisor: Senior Researcher Qiulan Huang</i>	2021/07 - 2021/08 Beijing, China
• Cooperate with developer and figure out the business process and accomplish documents about the workflow. • Implement Restful interfaces by Java for basic operations and rebuild the same workflow by Python.	

Computer Graphics & Spatial Acceleration

[Under Review] **H2-Ray: Hybrid and Hierarchical Ray Tracing for Efficient 3D Gaussian Rendering.**

T. Zhao, S. Li, Y. Chen, H. Ding, Z. Pan, C. Li.

[Under Review] **RTX-IPC: RT-Core-Accelerated Conservative Contact Search for IPC Simulation.**

T. Zhao, J. Guo, H. Zhao, C. Li.

Parallel Algorithms

[EDBT 2027] C2: Cache-Conscious Succinct Tries with Adaptive Unary Path Compression.

K. Zhang, **T. Zhao**, H. Xu.

[HPEC 2025] (**Best Student Paper Award**) pdGRASS: A Fast Parallel Density-Aware Algorithm for Graph Spectral Sparsification.

T. Zhao, Z. Yin, H. An, X. Yang, Z. Jin, J. Shen, H. Xu.

[Under Submission] Cache-Optimized Batch-Dynamic Succinct Bitvectors via Packed Memory Arrays.

S. Gao, **T. Zhao**, H. Komlós, L. Wang, H. Xu.

[Under Review] RabbitCD-HIT: Distributed High-Sensitivity Greedy Incremental Clustering for Billion-Scale Protein Sequences in Hours.

Y. Gao, G. Ma, Z. Yin, L. Yan, Y. Yang, T. Zhang, **T. Zhao**, Z. Liu, X. Duan, B. Niu, B. Schmidt, W. Liu.

AI & Systems (AI4Sys)

[AACL-IJCNLP 2026] LLM Agent Memory: A Survey from a Unified Representation–Management Perspective.

Z. Tang, X. He, **T. Zhao**, et al.

[AAAI 2026] OSVBench: Benchmarking LLMs on Specification Generation Tasks for Operating System Verification.

S. Li, J. Jiang, **T. Zhao**, J. Shen.

[Briefings in Bioinformatics 2024] ifDEEPre: Interpretable and Fast Deep Learning Model for Accurate Enzyme EC Number Predictions.

Q. Tan, J. Chen, Y. Wang, Z. Zhang, **T. Zhao**, and Y. Li.

[Under Review] A Multimodal Approach to Extract Page Transition Graphs from Mobile Applications.

S. Zhong, C. Liu, M. Ren, K. Chen, **T. Zhao**, J. Shen.

SELECTED AWARDS & SCHOLARSHIPS

Best Student Paper Award, IEEE HPEC 2025	09, 2025
Outstanding Graduate of Shandong Province, Government of Shandong Province	05, 2023
HPC China22: ACM/AMD International Parallel Computing Challenge, Second Prize (2/100+)	11, 2022
HPC China22: The 10th “Intel Cup” Parallel Application Challenge, Third Prize (4/100+)	11, 2022
ASC22 Student Supercomputer Challenge, Second Prize	02, 2022
National Scholarship, Ministry of Education of PRC	12, 2021
ICPC China National Invitational Programming Contest(Shannxi), Silver Medal	06, 2021
ICPC China Provincial Programming Contest(Shandong), Gold Medal	05, 2021
National Mathematics Competition (Non-mathematics Major), Second Prize	11, 2020

TEACHING EXPERIENCES

CSE8803 High Performance Algorithm Engineering	Georgia Tech, 2026 Spring
CSE6220 High Performance Computing	Georgia Tech, 2025 Spring
COMP2711 Discrete Mathematical Tools for Computer Science	HKUST, 2024 Spring